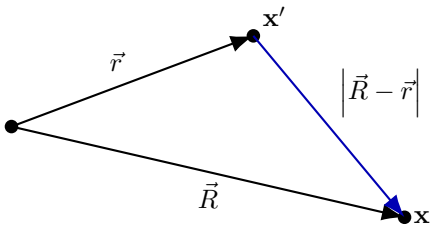
Suppose there exists two points with respective radius vectors $\vec{r}$ and $\vec{R}$ from the origin. Let $r < R$.



Consider the inverse of the distance between the points,

$$\begin{aligned}\frac{1}{\left|\vec{R}-\vec{r}\right|} &= \frac{1}{\sqrt{(\vec{R}-\vec{r})\cdot(\vec{R}-\vec{r})}} \\ &= \frac{1}{\sqrt{R^2-2Rr\cos\theta+r^2}} \\ &= \sum_{l=0}^{\infty}\frac{r^\ell}{R^{\ell+1}}P_\ell(\cos\theta) \qquad (1)\end{aligned}$$

This sum converges for $r < R$. $P_l(\cos\theta)$ are the Legndre polynomials with argument $\cos\theta$. The proof of the given identity [1] requires Cauchy's residue theorem and the Rodriguez formula for the Legendre polynomials. Now define a Coulomb potentital for some charge distribution. Assuming a neutral, closed shell atom, the distribution is then spherical.

An infinitesimal charge element within the distribution is given by $dq = \rho(\vec{r})\,d\tau$. The electrostatic potential at a fixed point $\vec{R}$ outside the charge distribution is obtained by summing the contributions from all charge elements throughout the volume of the atom. Here, $\vec{r}$ denotes the position of a charge element relative to the centre of the atom, while $\vec{R}$ denotes the position of the observation point relative to the same centre. The assumption that $r < R$ is maintained in the subsequent derivation of the dispersion interaction.

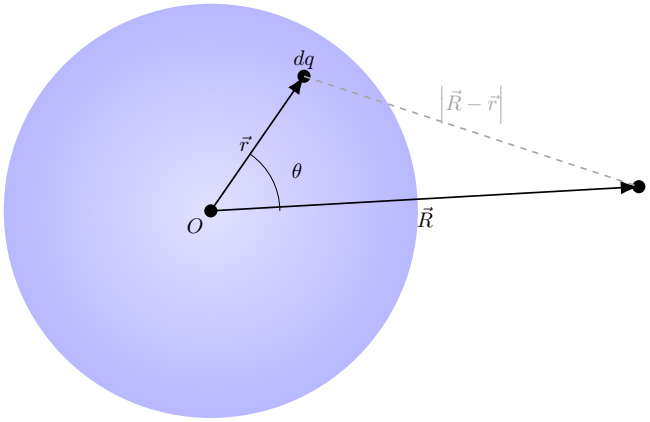


Figure 1: Geometry for the multipole expansion of the Coulomb potential due to a spherically symmetric charge distribution.

$$\begin{aligned} V(\vec{R}) &= \frac{1}{4\pi\epsilon_0}\iiint \frac{\rho(\vec{r})d\tau}{\left|\vec{R}-\vec{r}\right|} \\ &= \frac{1}{4\pi\epsilon_0}\iiint \rho(\vec{r})\left[\sum_{\ell=0}^{\infty}\frac{r^\ell}{R^{\ell+1}}P_\ell(\cos\theta)\right]d\tau \\ &= \frac{1}{4\pi\epsilon_0}\sum_{\ell=0}^{\infty}\frac{1}{R^{\ell+1}}\iiint r^\ell P_\ell(\cos\theta)\rho(\vec{r})d\tau \end{aligned} \tag{2}$$

Equation (2) separates the electrostatic potential into its multipole constituents. Specifically, $\ell = 0$ corresponds to the monopole contribution, $\ell = 1$ to the dipole contribution, $\ell = 2$ to the quadrupole contribution, and so on. We can now address the two body problem that constitutes the attractive dispersion force between two neutral atom's we call A and B.

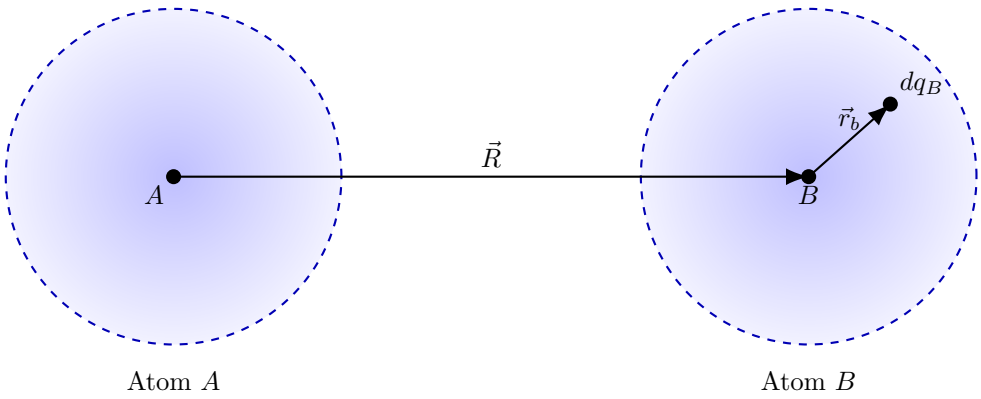


Figure 2: Geometry of two neutral atoms/ions with spherically symmetric electron clouds. The vector $\vec{R}$ joins the atomic centres, while $\vec{r}_b$ locates the volume element dq_B within atom B.

Atom A's electrostatic potential at a point $\vec{x}$ relative to its centre is represented by the multipole expansion

$$\phi_A(\vec{x}) = \frac{1}{4\pi\epsilon_0} \sum_{\ell=0}^{\infty} \frac{1}{x^{\ell+1}} \iiint r_A^{\ell} P_\ell(\vec{r}_A \cdot \vec{x}) \rho_A(\vec{r}_A) d^3 r_A. \tag{3}$$

An infinitesimal charge element in neutral atom B is given by

$$dq_B = \rho_B(\vec{r}_B)\, d^3 r_B.$$

Figure 2 depicts the electrostatic interaction between the multipole potential of atom A and the charge element dq_B. Since the position of dq_B relative to the centre of atom A is $\vec{x} = \vec{R} + \vec{r}_B$, its electrostatic interaction potential energy is

$$dU = \phi_A(\vec{R} + \vec{r}_B)\, dq_B.$$

This is the potential energy associated with placing the charge element in the electrostatic potential generated by atom A.

The total electrostatic interaction potential energy between atoms A and B is therefore

$$U_{AB}(\vec{R}) = \iiint \rho_B(\vec{r}_B) \phi_A(\vec{R} + \vec{r}_B)\, d^3 r_B. \tag{4}$$

For $r_B << R$ we can taylor expand ϕ_A. Note that, $\phi_{A,\alpha\beta} = \frac{\delta^2 \phi_A}{\delta R_\alpha \delta R_\beta}$. This mathematical short hand is utilised alongside Einstein summation to keep the expression succinct.

$$U_{AB} \sim \phi_A(\vec{R}) \underbrace{\iiint \rho_B(\vec{r}_B)\, d^3 r_B}_{\text{monopole, } q_B} + \phi_{A,\alpha}(\vec{R}) \underbrace{\iiint r_{B,\alpha} \rho_B(\vec{r}_B)\, d^3 r_B}_{\text{dipole, } \mu_{B,\alpha}} + \cdots \tag{5}$$

For neutural atoms, the monopole $q_B(\vec{r_B}) = 0$ trivally. Equation (3) gives the multipole expansion of the electrostatic potential generated by atom A. Since atom A is also neutral, its monopole term, $\ell = 0$, likewise vanishes. Thus, the leading non-zero contribution to the Van der Waals–London electrostatic interaction is the dipole–dipole term. This will hold true for the interaction between He-Li^+, as well as, He-F^-. To build intuition for the Van der Waals dispersion force, two simple harmonic oscillators can approximate the behaviour of the dipoles [2].

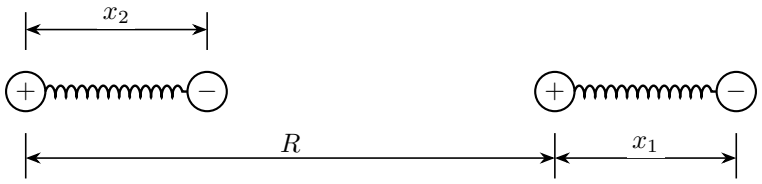


Figure 3: Model of the interaction between two fluctuating dipoles. Each neutral atom/ion is represented by a positive core and a negative charge coupled by a harmonic restoring force. The positive cores are separated by R, while x_1 and x_2 denote the instantaneous charge separations within the two atoms.

The Hamiltonian of the unperturbed system is given by [2]

$$H_0 = \frac{p_1^2}{2m} + \frac{1}{2} C x_1^2 + \frac{p_2^2}{2m} + \frac{1}{2} C x_2^2. \tag{6}$$

The constant $C = m w_0^2$, where w_0 is defined as the strongest optical absorption line of the atom. The intuition is that the the dipole–dipole interaction couples the ground states of the two atoms. As such, there will be a perturbing Coulombic interaction energy H_1 between the two oscillators. The geometry of the modelled interaction is shown in Figure 3. In CGS,

$$H_1 = \frac{e^2}{R} + \frac{e^2}{R + x_1 - x_2} - \frac{e^2}{R + x_1} - \frac{e^2}{R - x_2} \tag{7}$$

Working under the assumption of no electron cloud overlap, then

$$|x_1|,\ |x_2| \ll R.$$

To lowest order,

$$\mathcal{H}_1 \simeq -\frac{2e^2 x_1 x_2}{R^3}. \tag{8}$$

The problem is simplified by the normal mode transformation [2]

$$x_s \equiv \frac{1}{\sqrt{2}}(x_1 + x_2), \qquad x_a \equiv \frac{1}{\sqrt{2}}(x_1 - x_2). \tag{9}$$

Where the subscripts s and a denote symmetric and antisymmetric modes of motion. Furthermore we have momenta p_s and p_a associated with the two modes

$$p_s \equiv \frac{1}{\sqrt{2}}(p_1 + p_2), \qquad p_a \equiv \frac{1}{\sqrt{2}}(p_1 - p_2). \tag{10}$$

The total Hamiltonian $H_0 + H_1$ after the transformations (9) and (10) is

$$\mathcal{H} = \left[\frac{p_s^2}{2m} + \frac{1}{2}\left(C - \frac{2e^2}{R^3}\right)x_s^2\right] + \left[\frac{p_a^2}{2m} + \frac{1}{2}\left(C + \frac{2e^2}{R^3}\right)x_a^2\right]. \tag{11}$$

For an arbitrary harmonic oscillator

$$H = \frac{p^2}{2m} + \frac{1}{2}mw^2x^2.$$

Thus, we can conclude a symmetric oscillation frequency and an asymmetric oscillaton frequency.

$$w_s = \sqrt{\frac{C - 2e^2/R^3}{m}}, \qquad w_a = \sqrt{\frac{C + 2e^2/R^3}{m}}. \tag{12}$$

This leads to the conclusion that the angular frequency of the coupled harmonic system is given by

$$w = \underbrace{\sqrt{\frac{C}{m}}}_{w_0}\sqrt{1 \pm \frac{2e^2}{CR^3}}. \tag{13}$$

A binomial expansion can be applied to this expression for large R, such that $\frac{2e^2}{CR^3} \ll 1$. Therefore,

$$w \sim w_0\left(1 \pm \frac{e^2}{CR^3} - \frac{e^4}{2C^2R^6} + O(R^{-9})\right). \tag{14}$$

The only difference between w_s and w_a is in the first term where the $+$ indicates w_s and the $-$ indicates w_a.

The quantum harmonic oscillator has energy levels $E_n = \hbar w\left(n + \frac{1}{2}\right)$ [3]. The uncoupled ground-state energy is given by $2 \times \frac{1}{2}\hbar w_0 = \hbar w_0$. This holds for two neutral atoms separated by an infinite distance. In the limit $R \to \infty$, the multipole fields of each neutral atom vanish at the position of the other, and the two oscillators are uncoupled:

$$\lim_{R\to\infty} E_0(R) = \hbar w_0.$$

Subsequently, the interaction energy can be defined as the change in the ground-state energy when the atoms are brought from infinite separation to a distance R, so that their multipole fields couple:

$$\begin{aligned}
\Delta U(R) &= E_0(R) - \lim_{R\to\infty} E_0(R) \\
&= \frac{1}{2}\hbar(w_s + w_a) - \hbar w_0 \\
&\approx \frac{\hbar w_0}{2}\left[2 - \frac{1}{4}\left(\frac{2e^2}{CR^3}\right)^2\right] - \hbar w_0 \\
&= -\frac{\hbar w_0 e^4}{2C^2R^6} \\
&= -\frac{C_6}{R^6}, \qquad (15)
\end{aligned}$$

where

$$C_6 = \frac{\hbar w_0 e^4}{2C^2}. \qquad (16)$$

The minus sign in Eq. (15) indicates an attractive interaction whose magnitude varies as the inverse sixth power of the separation between the two oscillators. This is the van der Waals–London interaction. It is the principal interaction in crystals of inert gases and also in crystals of many organic molecules [2]. The fluctuating dipole of one atom couples to and polarises the other, mixing excited-state configurations into the joint ground state of the two-atom system [4, 5]. To relate the restoring constant C in Eq. (6) to the atomic polarisability, consider a harmonically bound electron in a static electric field E. The restoring force has magnitude Cx, while the electric force has magnitude eE. At equilibrium, these forces balance, giving $Cx = eE$. The induced dipole moment has magnitude $p = ex$, and since $p = \alpha E$, the static dipole polarisability

$$\alpha = \frac{e^2}{C}.$$

Equation (16) may be written as

$$C_6 = \frac{1}{2}\hbar w_0 \alpha^2,$$

showing that, within this simplified model, C_6 is proportional to the characteristic excitation frequency w_0 and to the square of the atomic polarisability. The coupling lowers the ground-state energy relative to that of the two isolated atoms producing an attractive force. The simplified system assumes that the two neutral atoms are of the same species and consequently have the same characteristic optical absorption frequency w_0. More formal treatments of interactions between dissimilar neutral atoms and ions have been conducted in the literature [4, 5, 6, 7, 8]. In the formal treatment requiring pertubation theories, the long-range interaction between two spherically symmetric atoms can be written in the general form

$$\Delta U(R) \sim -\frac{C_6}{R^6} - \frac{C_8}{R^8} - \frac{C_{10}}{R^{10}} - \cdots = \sum_{n=3}^{\infty} -\frac{C_{2n}}{R^{2n}}. \tag{17}$$

Using second-order perturbation theory, Visentin [4] derives the following dispersion coefficients for two dissimilar species A and B:

$$C_6^{AB} = \frac{3}{\pi}\int_0^{\infty} \alpha_1^A(i\omega)\alpha_1^B(i\omega)\, d\omega, \tag{18}$$

$$C_8^{AB} = \frac{15}{2\pi}\int_0^{\infty} \left[\alpha_1^A(i\omega)\alpha_2^B(i\omega) + \alpha_2^A(i\omega)\alpha_1^B(i\omega)\right] d\omega, \tag{19}$$

$$C_{10}^{AB} = \frac{14}{\pi}\int_0^{\infty} \left[\alpha_1^A(i\omega)\alpha_3^B(i\omega) + \alpha_3^A(i\omega)\alpha_1^B(i\omega)\right] d\omega + \frac{35}{\pi}\int_0^{\infty} \alpha_2^A(i\omega)\alpha_2^B(i\omega)\, d\omega. \tag{20}$$

Here, C_6 describes the dipole–dipole interaction, C_8 describes the dipole–quadrupole interaction, and C_{10} contains contributions from both dipole–octupole and quadrupole–quadrupole interactions. Furthermore, α_1, α_2, and α_3 denote the dipole, quadrupole, and octupole polarisabilities, respectively. More generally, α_n denotes the polarisability associated with the multipole operator $\hat{Q}_n$ of order n.

$$\alpha_n(iu) = 2\sum_k \frac{(E_k - E_0)\left|\langle 0|\hat{Q}_n|k\rangle\right|^2}{(E_k - E_0)^2 + \hbar^2 u^2}. \tag{21}$$

The dynamic multipole polarisability describes how the applied field couples the ground state to excited states through the multipole operator $\hat{Q}_n$. This coupling is captured by the term

$$\left|\langle 0|\hat{Q}_n|k\rangle\right|^2,$$

which gives the transition strength between the ground state $|0\rangle$ and the kth excited state $|k\rangle$. Each transition is weighted by its excitation energy $E_k - E_0$ and by the imaginary-frequency variable u. The summation over all excited

states incorporates the full excitation spectrum and therefore provides a rigorous extension of the simplified dipole–dipole harmonic-oscillator model, in which the atom was represented by a single characteristic absorption frequency. The more strongly the multipole operator $\hat{Q}_n$ couples the ground state to the excited states, the larger the multipole polarisability α_n. An important takeaway is that higher-order multipole interactions decay more rapidly as R increases. In many situations, the dipole–dipole interaction is sufficient, and the C_6 coefficient provides the dominant long-range contribution [2]. In the limit $R \to \infty$, all multipole interaction terms vanish.

Bibliography

[1] G. B. Arfken, H. J. Weber, and F. E. Harris, *Mathematical Methods for Physicists: A Comprehensive Guide*, 7th ed. (Academic Press, Waltham, MA, 2013).

[2] C. Kittel, *Introduction to Solid State Physics*, 8th ed. (John Wiley & Sons, Hoboken, NJ, 2005).

[3] R. Shankar, *Principles of Quantum Mechanics*, 2nd ed., Springer, New York, 1994.

[4] G. Visentin, *Accurate Ab Initio Evaluation of the Interatomic Potentials and Long-Range Coefficients*, Ph.D. thesis, Skolkovo Institute of Science and Technology, Moscow, Russia (2021).

[5] A. Dalgarno and J. T. Lewis, "The exact calculation of long-range forces between atoms by perturbation theory," *Proceedings of the Royal Society of London. Series A, Mathematical and Physical Sciences*, **233**, 70–74 (1955). doi:10.1098/rspa.1955.0246.

[6] A. Dalgarno and W. D. Davison, "Long-range interactions of alkali metals," *Molecular Physics*, **13**, 479–486 (1967). doi:10.1080/00268976700101371.

[7] S. Singh, K. Kaur, B. K. Sahoo, and B. Arora, "Dispersion coefficients for the interaction of inert gas atoms with alkali and alkaline earth ions and alkali atoms with their singly ionized ions," *Asian Journal of Physics*, **25**, 655–666 (2016).

[8] J. Jiang, J. Mitroy, Y. Cheng, and M. W. J. Bromley, "Effective oscillator strength distributions of spherically symmetric atoms for calculating polarizabilities and long-range atom–atom interactions," *Atomic Data and Nuclear Data Tables*, **101**, 158–186 (2015). doi:10.1016/j.adt.2014.10.001.